

1. (12) Learn the definition of P-complete (Arora Textbook def 6.28). Show that if L is P-complete then $L \in \text{NC}$ iff $\text{P} = (\text{uniform}) \text{NC}$
2. (24) show that the majority function cannot be computed in AC^0 .
The majority function $\text{maj} : \{0, 1\}^n \rightarrow \{0, 1\}$: outputs 1 if the number of 1s in the input is at least $n/2$; outputs 0 otherwise.
3. (20) Show that one can efficiently simulate choosing a random number from 1 to N (denote this set as $[N]$) using coin tosses. That is, show that for every N and $\delta > 0$, there is a probabilistic algorithm A running in $\text{poly}(\log N \log(1/\delta))$ time with output in $\{1, 2, \dots, N, ?\}$ such that
 - (a) conditioned on not outputting $?$, A 's output is uniformly distributed in $[N]$ and
 - (b) the probability that A outputs $?$ is at most δ .
4. (22) Learn the definition of randomized polynomial time reduction and the definition of $\text{BP}\cdot\text{NP}$ in Arora Textbook section 7.6. and solve the following problem.
A nondeterministic circuit C has two inputs x, y . We say that C accepts x iff there exists y such that $C(x, y) = 1$. The size of the circuit is measured as a function of $|x|$. Let NP/poly be the languages that are decided by polynomial size nondeterministic circuits. Show that $\text{BP}\cdot\text{NP} \subseteq \text{NP/poly}$.
5. (22) Show that $\text{BPL} \subseteq \text{P}$.